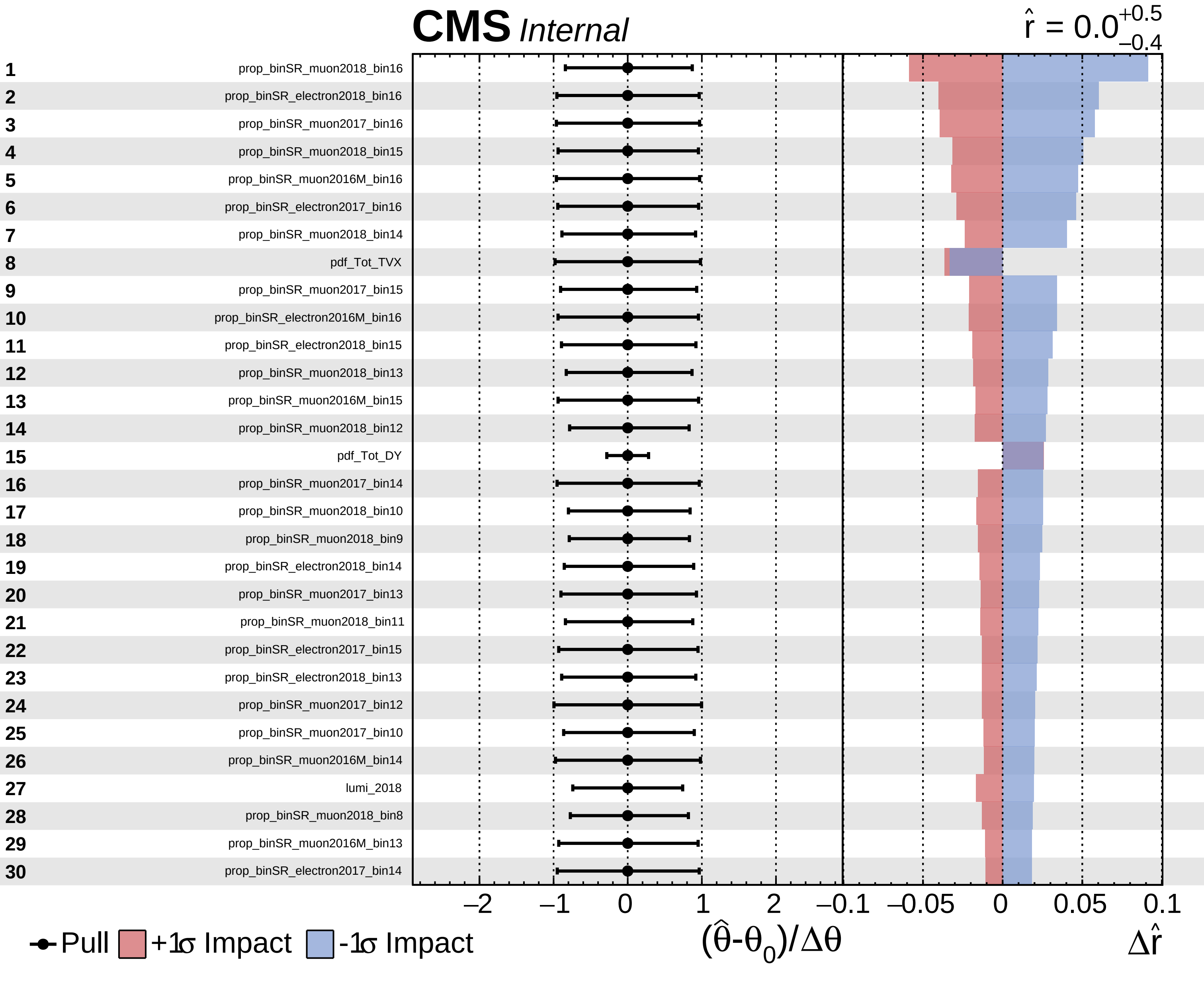
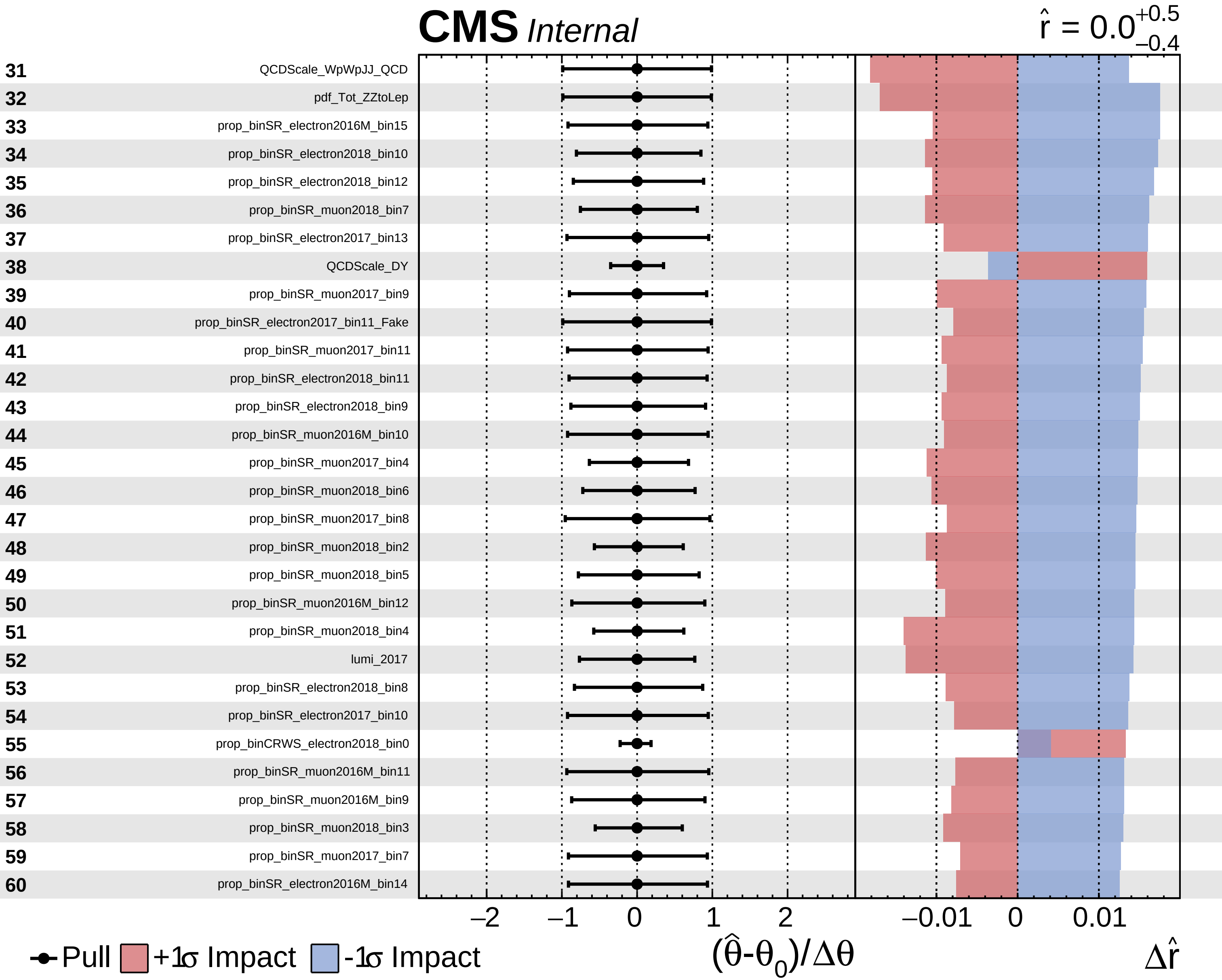
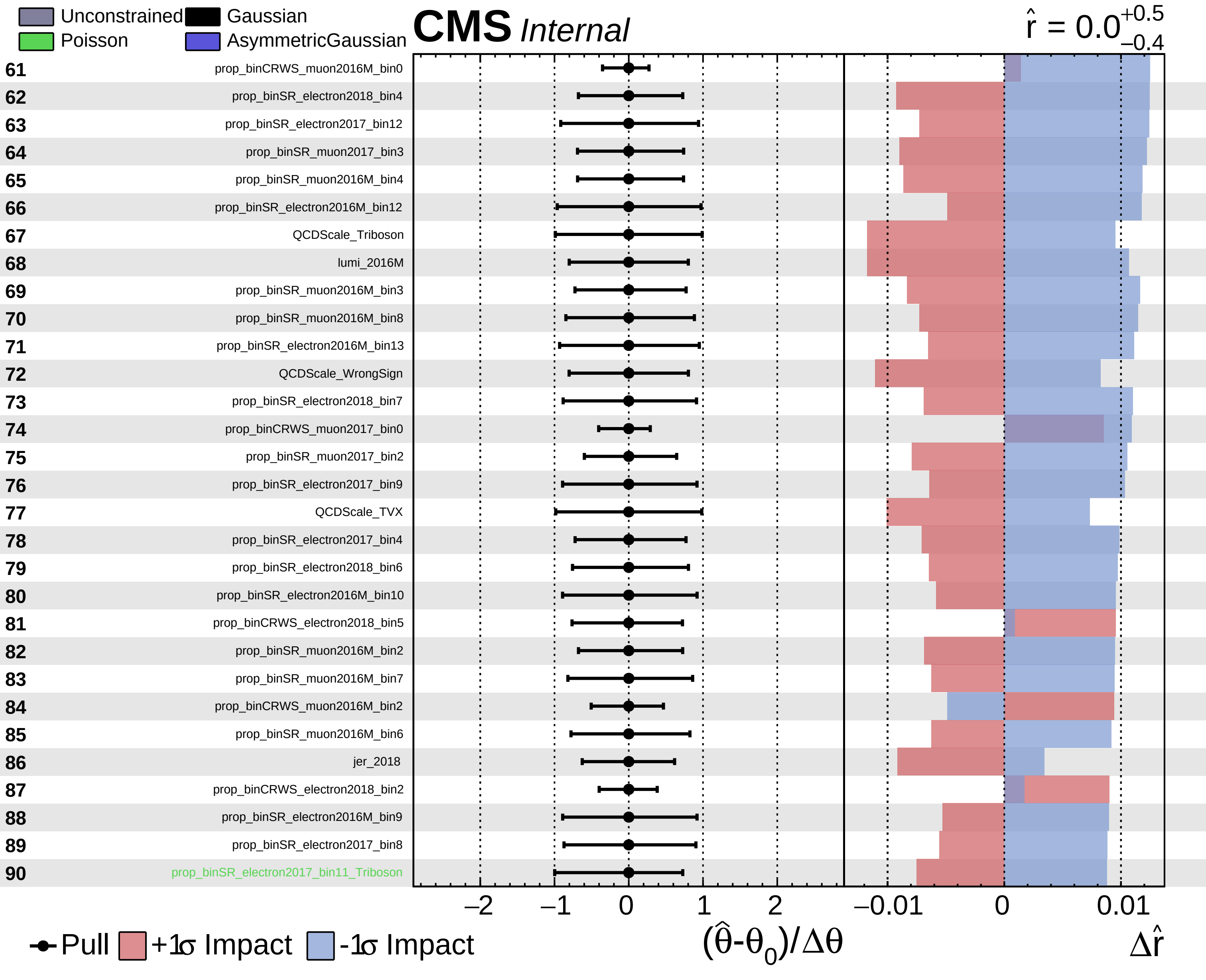
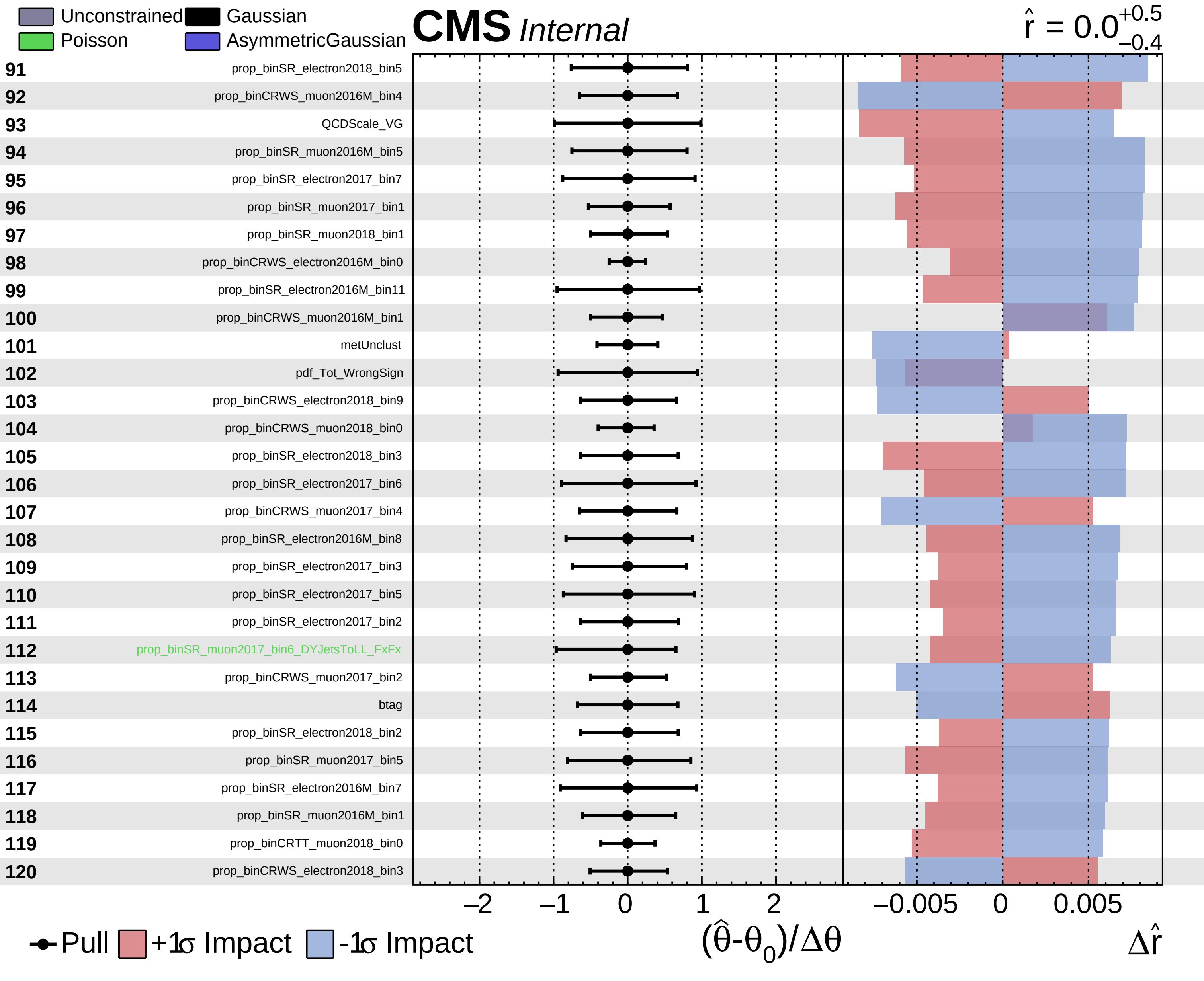

CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
1 prop_binSR_muon2018_bin16
2 prop_binSR_electron2018_bin16
3 prop_binSR_muon2017_bin16
4 prop_binSR_muon2018_bin15
5 prop_binSR_muon2016M_bin16
6 prop_binSR_electron2017_bin16
7 prop_binSR_muon2018_bin14
8 pdf_Tot_TVX
9 prop_binSR_muon2017_bin15
10 prop_binSR_electron2016M_bin16
11 prop_binSR_electron2018_bin15
12 prop_binSR_muon2018_bin13
13 prop_binSR_muon2016M_bin15
14 prop_binSR_muon2018_bin12
15 pdf_Tot_DY
16 prop_binSR_muon2017_bin14
17 prop_binSR_muon2018_bin10
18 prop_binSR_muon2018_bin9
19 prop_binSR_electron2018_bin14
20 prop_binSR_muon2017_bin13
21 prop_binSR_muon2018_bin11
22 prop_binSR_electron2017_bin15
23 prop_binSR_electron2018_bin13
24 prop_binSR_muon2017_bin12
25 prop_binSR_muon2017_bin10
26 prop_binSR_muon2016M_bin14
27 lumi_2018
28 prop_binSR_muon2018_bin8
29 prop_binSR_muon2016M_bin13
30 prop_binSR_electron2017_bin14
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



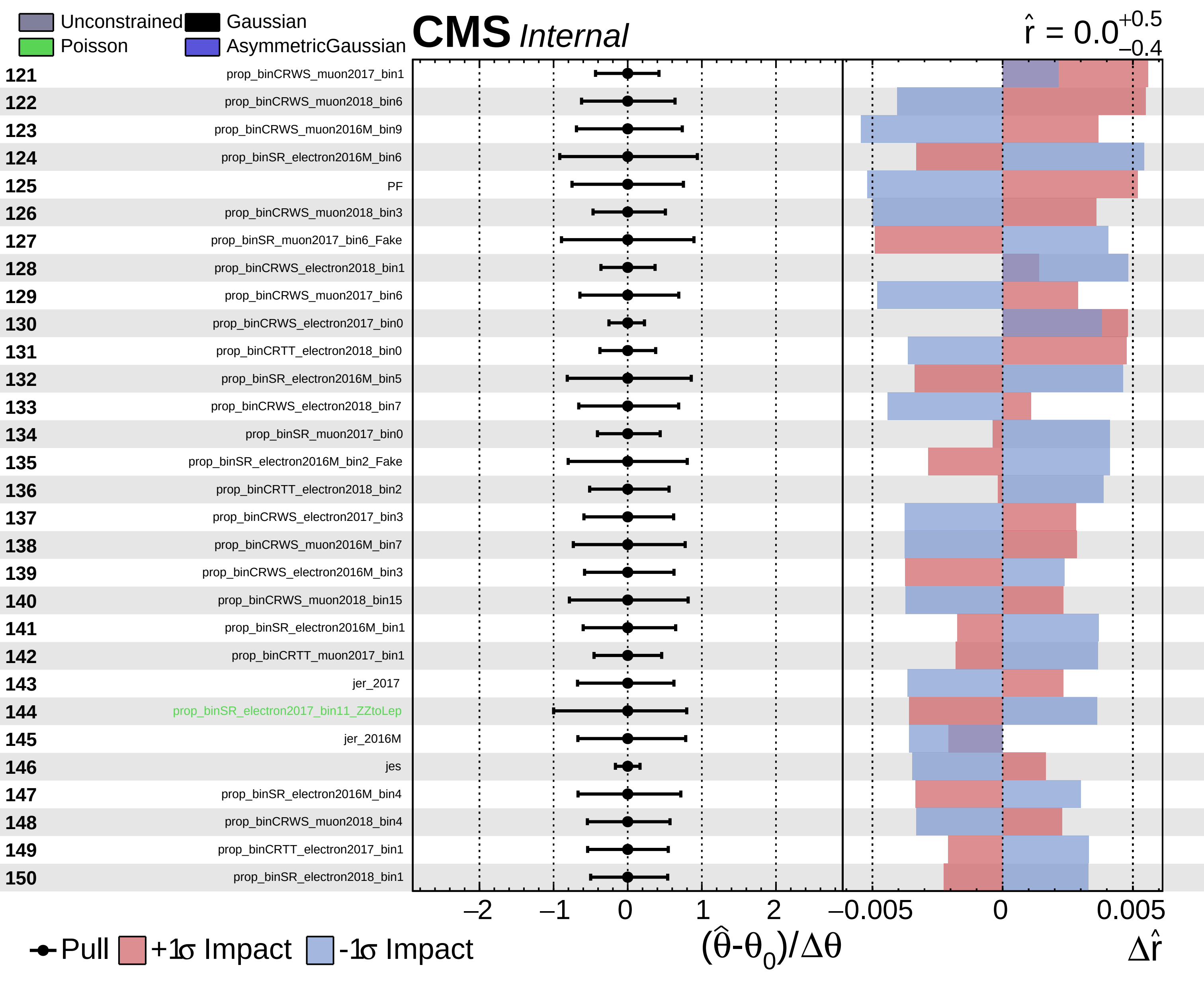
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
31 QCDScale_WpWpJJ_QCD
32 pdf_Tot_ZZtoLep
33 prop_binSR_electron2016M_bin15
34 prop_binSR_electron2018_bin10
35 prop_binSR_electron2018_bin12
36 prop_binSR_muon2018_bin7
37 prop_binSR_electron2017_bin13
38 QCDScale_DY
39 prop_binSR_muon2017_bin9
40 prop_binSR_electron2017_bin11_Fake
41 prop_binSR_muon2017_bin11
42 prop_binSR_electron2018_bin11
43 prop_binSR_electron2018_bin9
44 prop_binSR_muon2016M_bin10
45 prop_binSR_muon2017_bin4
46 prop_binSR_muon2018_bin6
47 prop_binSR_muon2017_bin8
48 prop_binSR_muon2018_bin2
49 prop_binSR_muon2018_bin5
50 prop_binSR_muon2016M_bin12
51 prop_binSR_muon2018_bin4
52 lumi_2017
53 prop_binSR_electron2018_bin8
54 prop_binSR_electron2017_bin10
55 prop_binCRWS_electron2018_bin0
56 prop_binSR_muon2016M_bin11
57 prop_binSR_muon2016M_bin9
58 prop_binSR_muon2018_bin3
59 prop_binSR_muon2017_bin7
60 prop_binSR_electron2016M_bin14
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



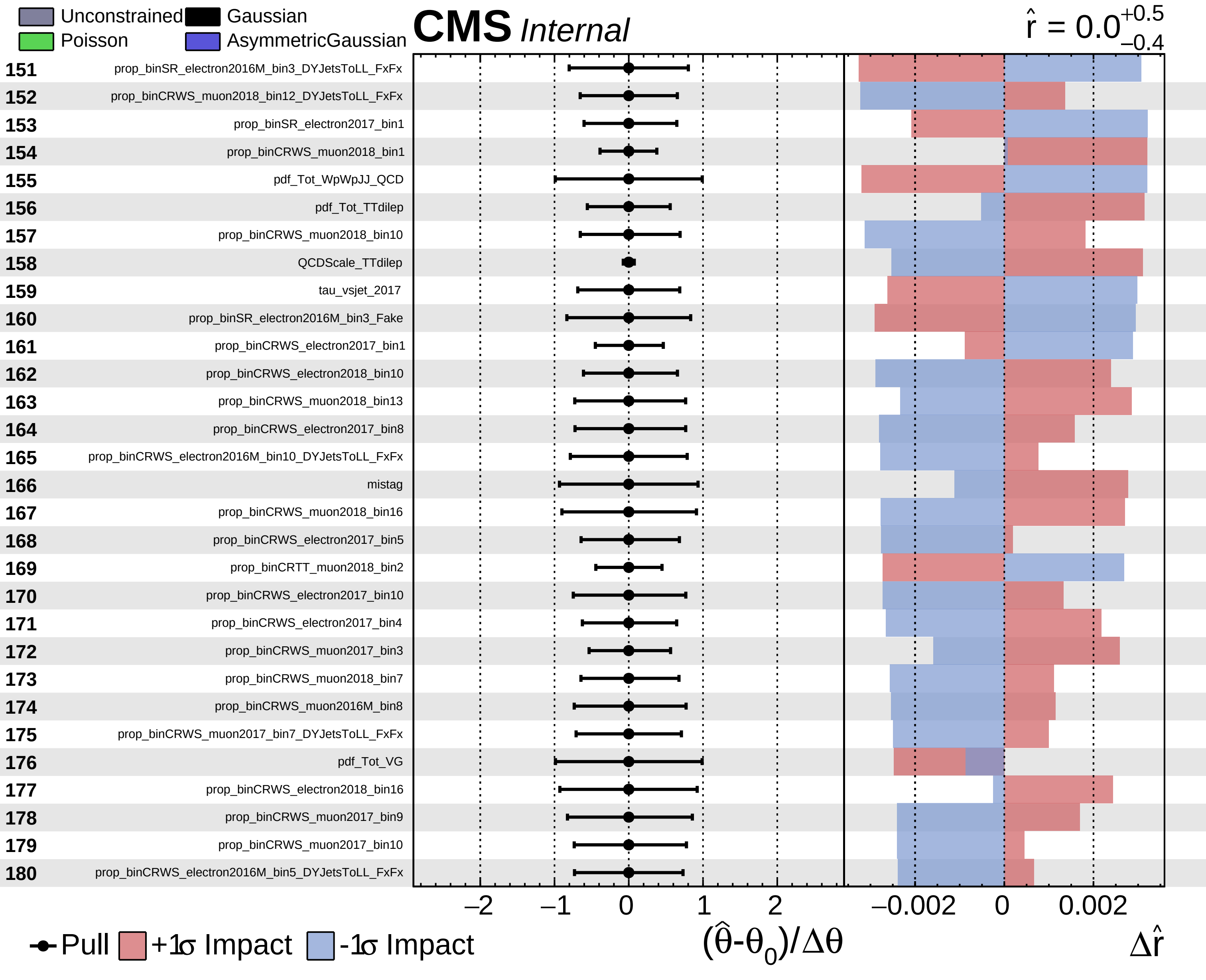
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
61 prop_binCRWS_muon2016M_bin0
62 prop_binSR_electron2018_bin4
63 prop_binSR_electron2017_bin12
64 prop_binSR_muon2017_bin3
65 prop_binSR_muon2016M_bin4
66 prop_binSR_electron2016M_bin12
67 QCDScale_Triboson
68 lumi_2016M
69 prop_binSR_muon2016M_bin3
70 prop_binSR_muon2016M_bin8
71 prop_binSR_electron2016M_bin13
72 QCDScale_WrongSign
73 prop_binSR_electron2018_bin7
74 prop_binCRWS_muon2017_bin0
75 prop_binSR_muon2017_bin2
76 prop_binSR_electron2017_bin9
77 QCDScale_TVX
78 prop_binSR_electron2017_bin4
79 prop_binSR_electron2018_bin6
80 prop_binSR_electron2016M_bin10
81 prop_binCRWS_electron2018_bin5
82 prop_binSR_muon2016M_bin2
83 prop_binSR_muon2016M_bin7
84 prop_binCRWS_muon2016M_bin2
85 prop_binSR_muon2016M_bin6
86 jer_2018
87 prop_binCRWS_electron2018_bin2
88 prop_binSR_electron2016M_bin9
89 prop_binSR_electron2017_bin8
90 prop_binSR_electron2017_bin11_Triboson
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

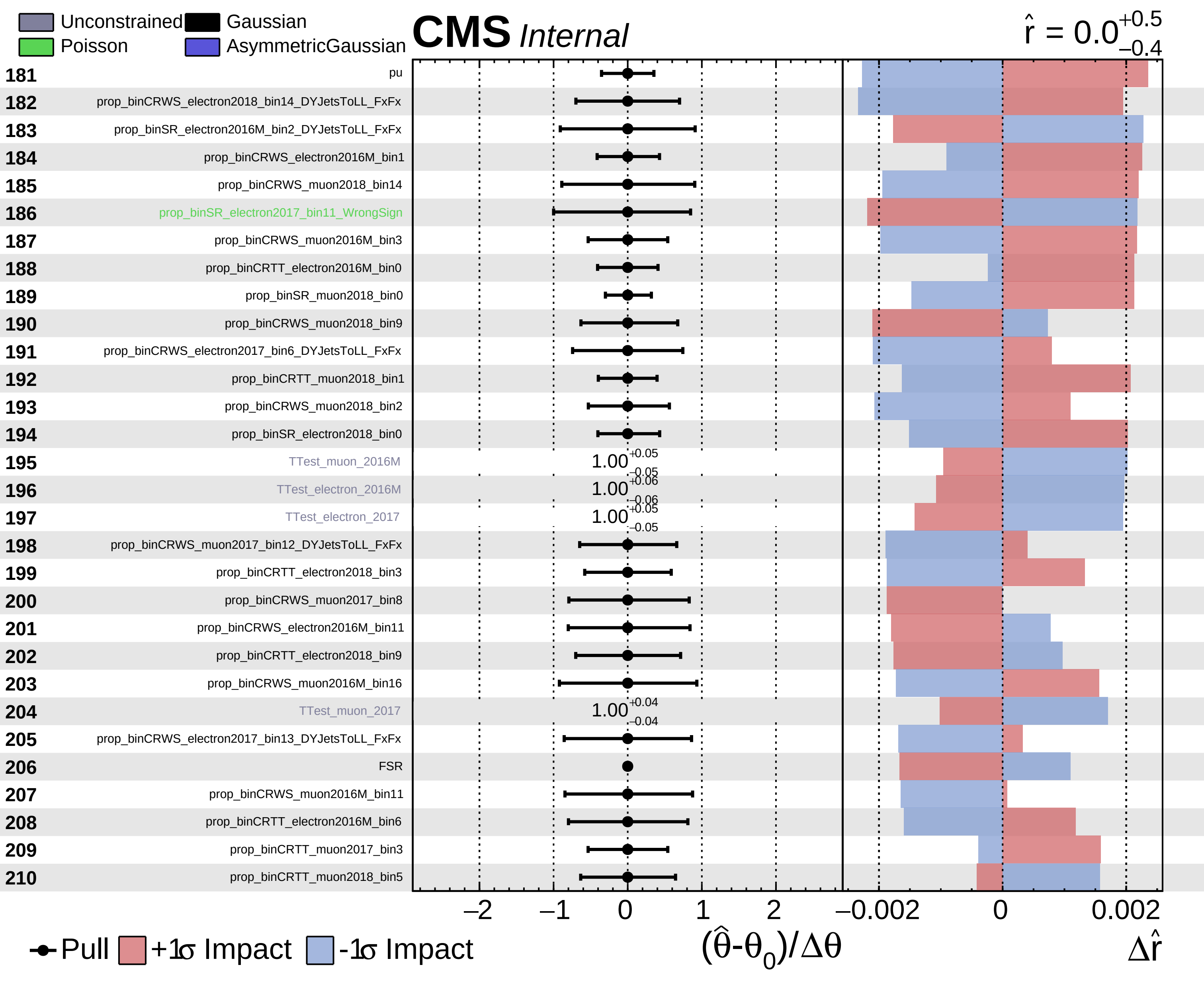


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
91 prop_binSR_electron2018_bin5
92 prop_binCRWS_muon2016M_bin4
93 QCDScale_VG
94 prop_binSR_muon2016M_bin5
95 prop_binSR_electron2017_bin7
96 prop_binSR_muon2017_bin1
97 prop_binSR_muon2018_bin1
98 prop_binCRWS_electron2016M_bin0
99 prop_binSR_electron2016M_bin11
100 prop_binCRWS_muon2016M_bin1
101 metUnclust
102 pdf_Tot_WrongSign
103 prop_binCRWS_electron2018_bin9
104 prop_binCRWS_muon2018_bin0
105 prop_binSR_electron2018_bin3
106 prop_binSR_electron2017_bin6
107 prop_binCRWS_muon2017_bin4
108 prop_binSR_electron2016M_bin8
109 prop_binSR_electron2017_bin3
110 prop_binSR_electron2017_bin5
111 prop_binSR_electron2017_bin2
112 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
113 prop_binCRWS_muon2017_bin2
114 btag
115 prop_binSR_electron2018_bin2
116 prop_binSR_muon2017_bin5
117 prop_binSR_electron2016M_bin7
118 prop_binSR_muon2016M_bin1
119 prop_binCRTT_muon2018_bin0
120 prop_binCRWS_electron2018_bin3
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

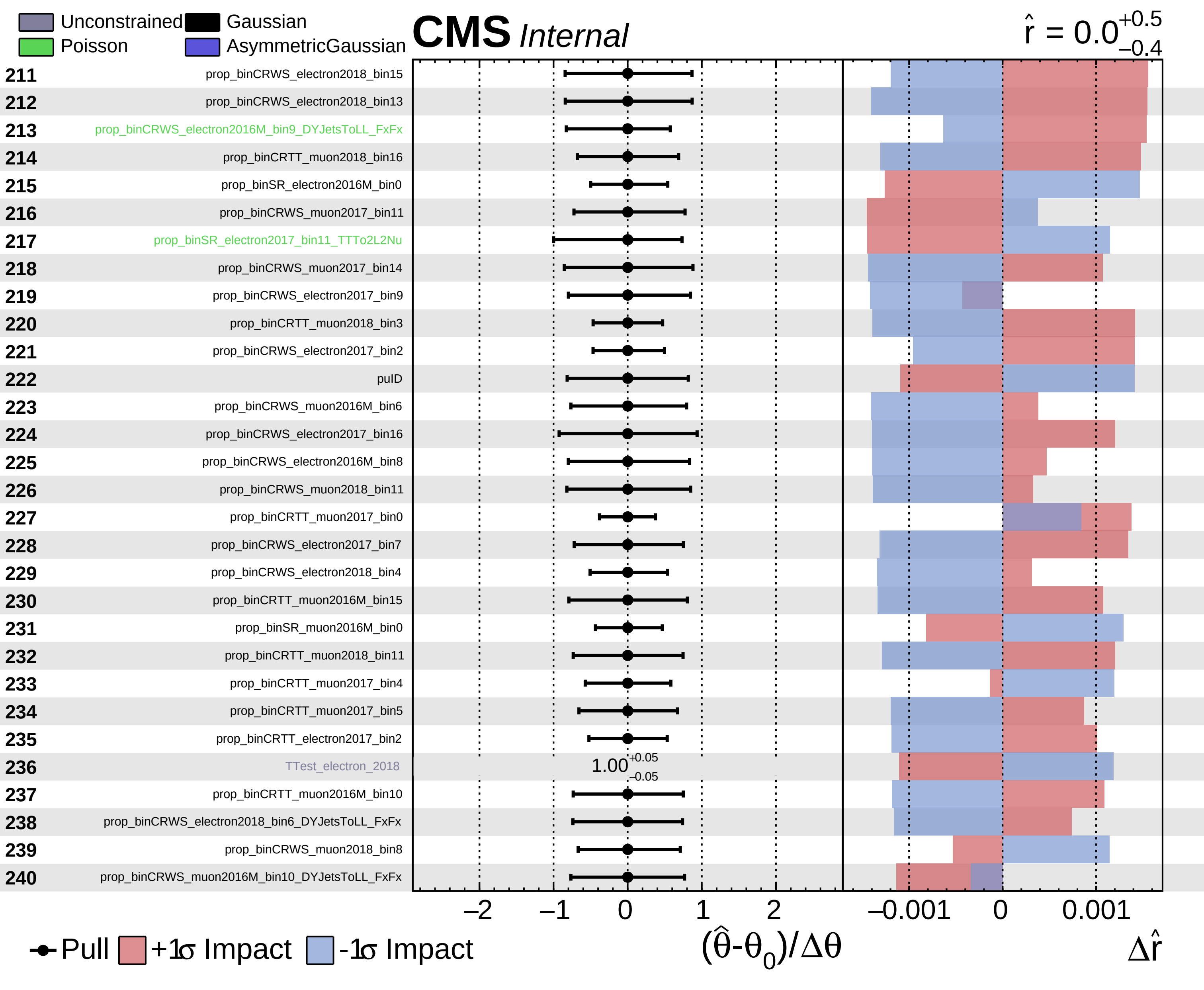


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
121 prop_binCRWS_muon2017_bin1
122 prop_binCRWS_muon2018_bin6
123 prop_binCRWS_muon2016M_bin9
124 prop_binSR_electron2016M_bin6
125 PF
126 prop_binCRWS_muon2018_bin3
127 prop_binSR_muon2017_bin6_Fake
128 prop_binCRWS_electron2018_bin1
129 prop_binCRWS_muon2017_bin6
130 prop_binCRWS_electron2017_bin0
131 prop_binCRTT_electron2018_bin0
132 prop_binSR_electron2016M_bin5
133 prop_binCRWS_electron2018_bin7
134 prop_binSR_muon2017_bin0
135 prop_binSR_electron2016M_bin2_Fake
136 prop_binCRTT_electron2018_bin2
137 prop_binCRWS_electron2017_bin3
138 prop_binCRWS_muon2016M_bin7
139 prop_binCRWS_electron2016M_bin3
140 prop_binCRWS_muon2018_bin15
141 prop_binSR_electron2016M_bin1
142 prop_binCRTT_muon2017_bin1
143 jer_2017
144 prop_binSR_electron2017_bin11_ZZtoLep
145 jer_2016M
146 jes
147 prop_binSR_electron2016M_bin4
148 prop_binCRWS_muon2018_bin4
149 prop_binCRTT_electron2017_bin1
150 prop_binSR_electron2018_bin1
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

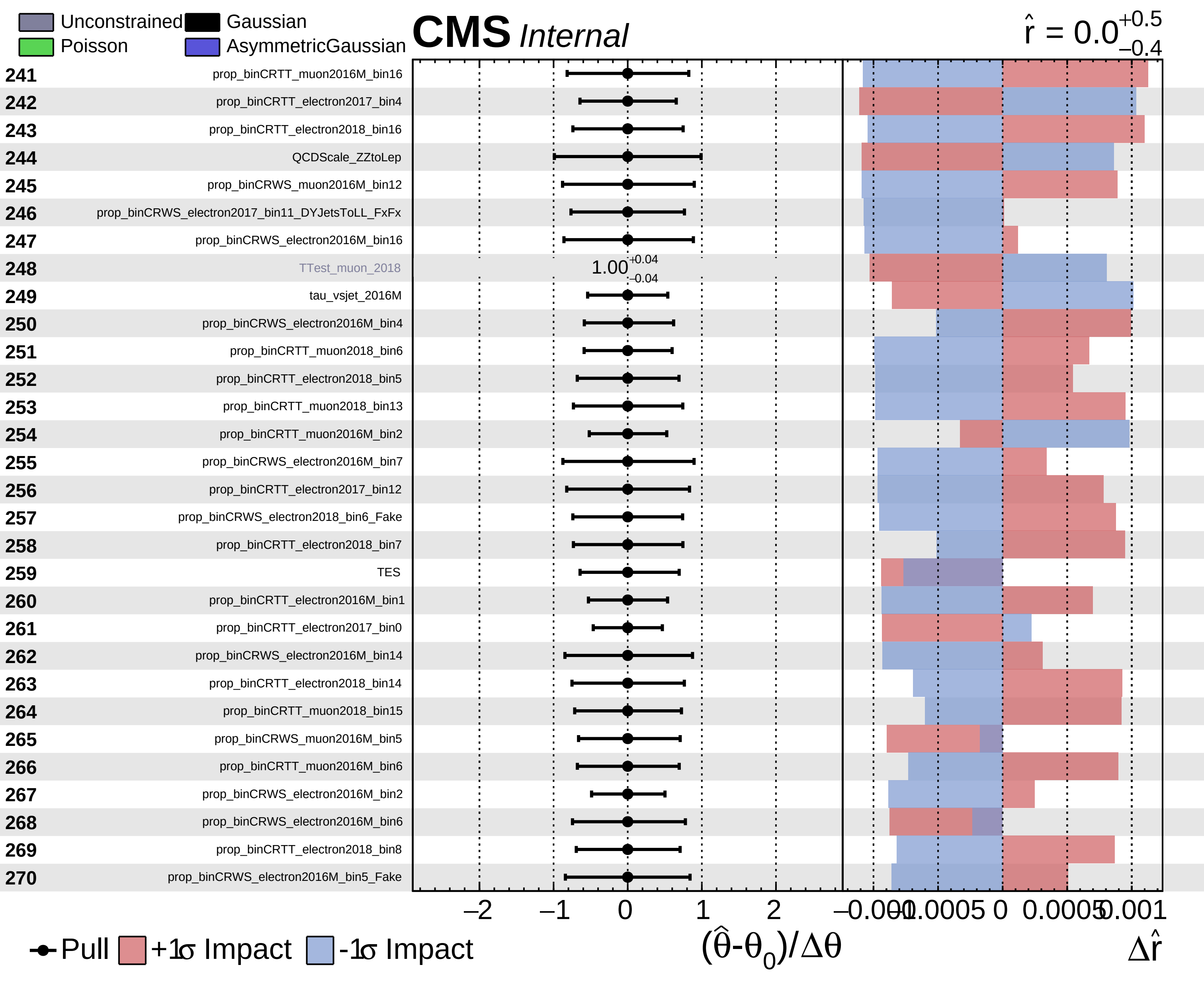

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
151 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
152 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
153 prop_binSR_electron2017_bin1
154 prop_binCRWS_muon2018_bin1
155 pdf_Tot_WpWpJJ_QCD
156 pdf_Tot_TTdilep
157 prop_binCRWS_muon2018_bin10
158 QCDScale_TTdilep
159 tau_vsjet_2017
160 prop_binSR_electron2016M_bin3_Fake
161 prop_binCRWS_electron2017_bin1
162 prop_binCRWS_electron2018_bin10
163 prop_binCRWS_muon2018_bin13
164 prop_binCRWS_electron2017_bin8
165 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
166 mistag
167 prop_binCRWS_muon2018_bin16
168 prop_binCRWS_electron2017_bin5
169 prop_binCRTT_muon2018_bin2
170 prop_binCRWS_electron2017_bin10
171 prop_binCRWS_electron2017_bin4
172 prop_binCRWS_muon2017_bin3
173 prop_binCRWS_muon2018_bin7
174 prop_binCRWS_muon2016M_bin8
175 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
176 pdf_Tot_VG
177 prop_binCRWS_electron2018_bin16
178 prop_binCRWS_muon2017_bin9
179 prop_binCRWS_muon2017_bin10
180 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



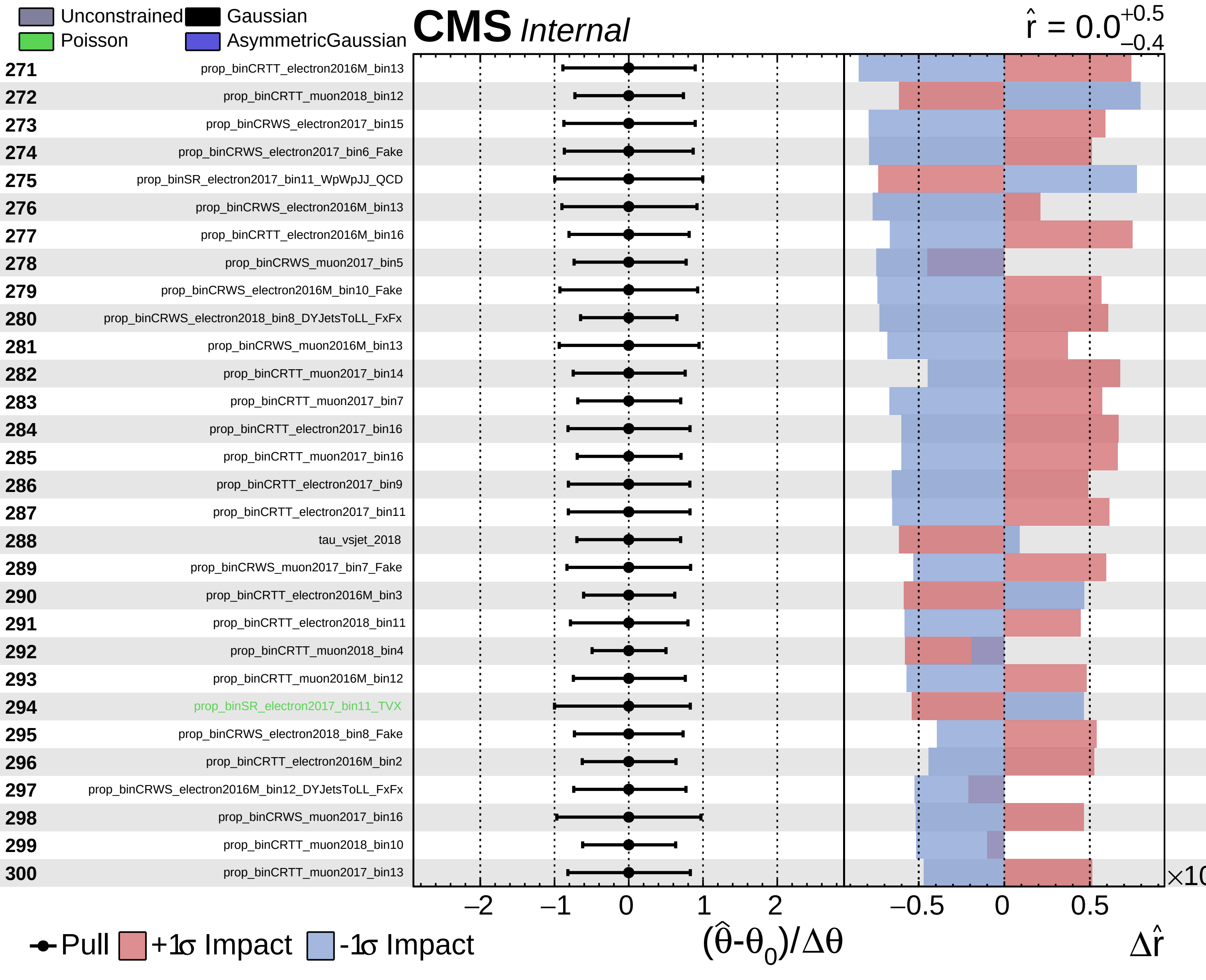
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
181 pu
182 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
183 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
184 prop_binCRWS_electron2016M_bin1
185 prop_binCRWS_muon2018_bin14
186 prop_binSR_electron2017_bin11_WrongSign
187 prop_binCRWS_muon2016M_bin3
188 prop_binCRTT_electron2016M_bin0
189 prop_binSR_muon2018_bin0
190 prop_binCRWS_muon2018_bin9
191 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
192 prop_binCRTT_muon2018_bin1
193 prop_binCRWS_muon2018_bin2
194 prop_binSR_electron2018_bin0
195 TTest_muon_2016M $1.00^{+0.05}_{-0.05}$
196 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
197 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
198 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
199 prop_binCRTT_electron2018_bin3
200 prop_binCRWS_muon2017_bin8
201 prop_binCRWS_electron2016M_bin11
202 prop_binCRTT_electron2018_bin9
203 prop_binCRWS_muon2016M_bin16
204 TTest_muon_2017 $1.00^{+0.04}_{-0.04}$
205 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
206 FSR
207 prop_binCRWS_muon2016M_bin11
208 prop_binCRTT_electron2016M_bin6
209 prop_binCRTT_muon2017_bin3
210 prop_binCRTT_muon2018_bin5
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



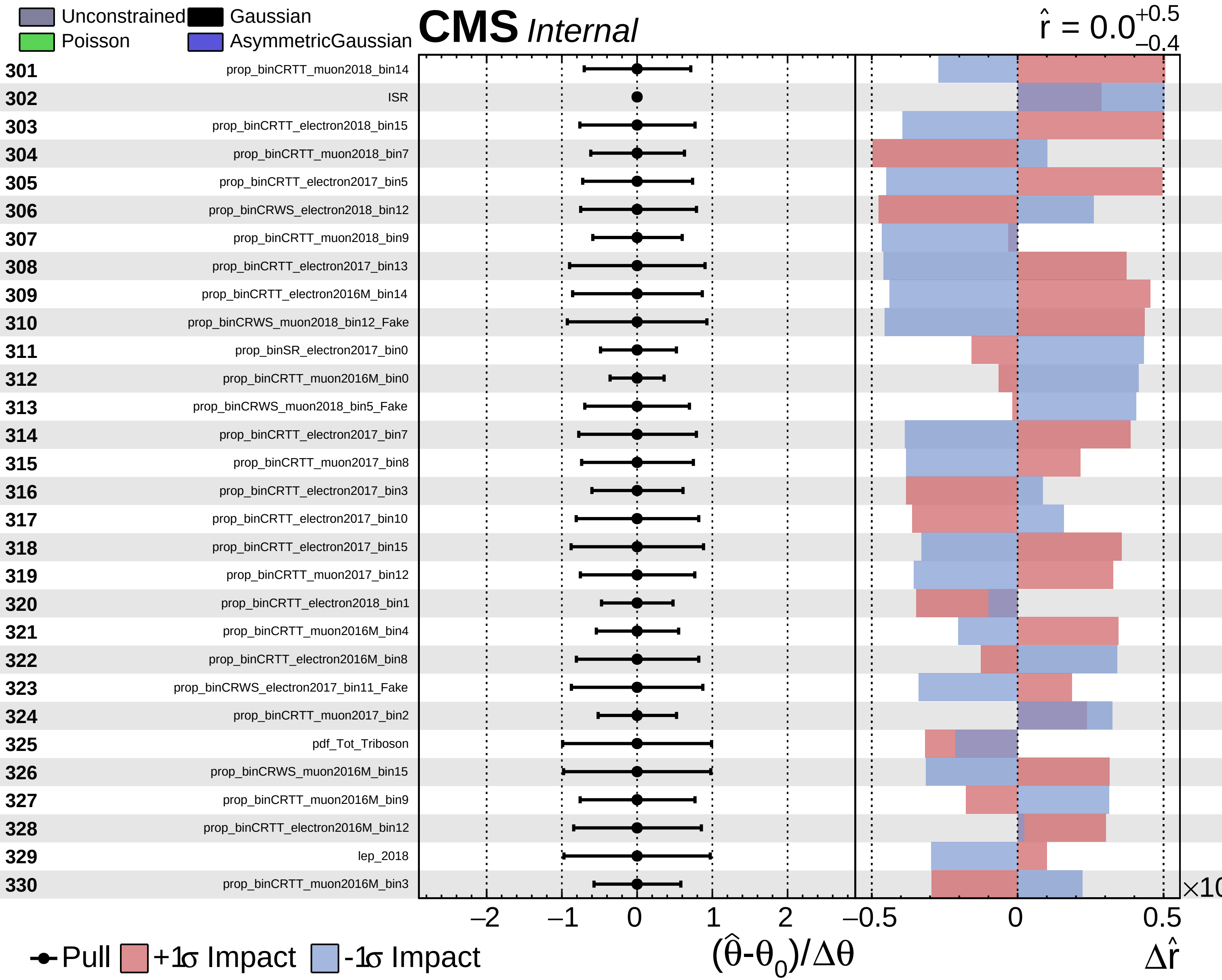
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
211 prop_binCRWS_electron2018_bin15
212 prop_binCRWS_electron2018_bin13
213 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
214 prop_binCRTT_muon2018_bin16
215 prop_binSR_electron2016M_bin0
216 prop_binCRWS_muon2017_bin11
217 prop_binSR_electron2017_bin11_TTTo2L2Nu
218 prop_binCRWS_muon2017_bin14
219 prop_binCRWS_electron2017_bin9
220 prop_binCRTT_muon2018_bin3
221 prop_binCRWS_electron2017_bin2
222 puID
223 prop_binCRWS_muon2016M_bin6
224 prop_binCRWS_electron2017_bin16
225 prop_binCRWS_electron2016M_bin8
226 prop_binCRWS_muon2018_bin11
227 prop_binCRTT_muon2017_bin0
228 prop_binCRWS_electron2017_bin7
229 prop_binCRWS_electron2018_bin4
230 prop_binCRTT_muon2016M_bin15
231 prop_binSR_muon2016M_bin0
232 prop_binCRTT_muon2018_bin11
233 prop_binCRTT_muon2017_bin4
234 prop_binCRTT_muon2017_bin5
235 prop_binCRTT_electron2017_bin2
236 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
237 prop_binCRTT_muon2016M_bin10
238 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
239 prop_binCRWS_muon2018_bin8
240 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



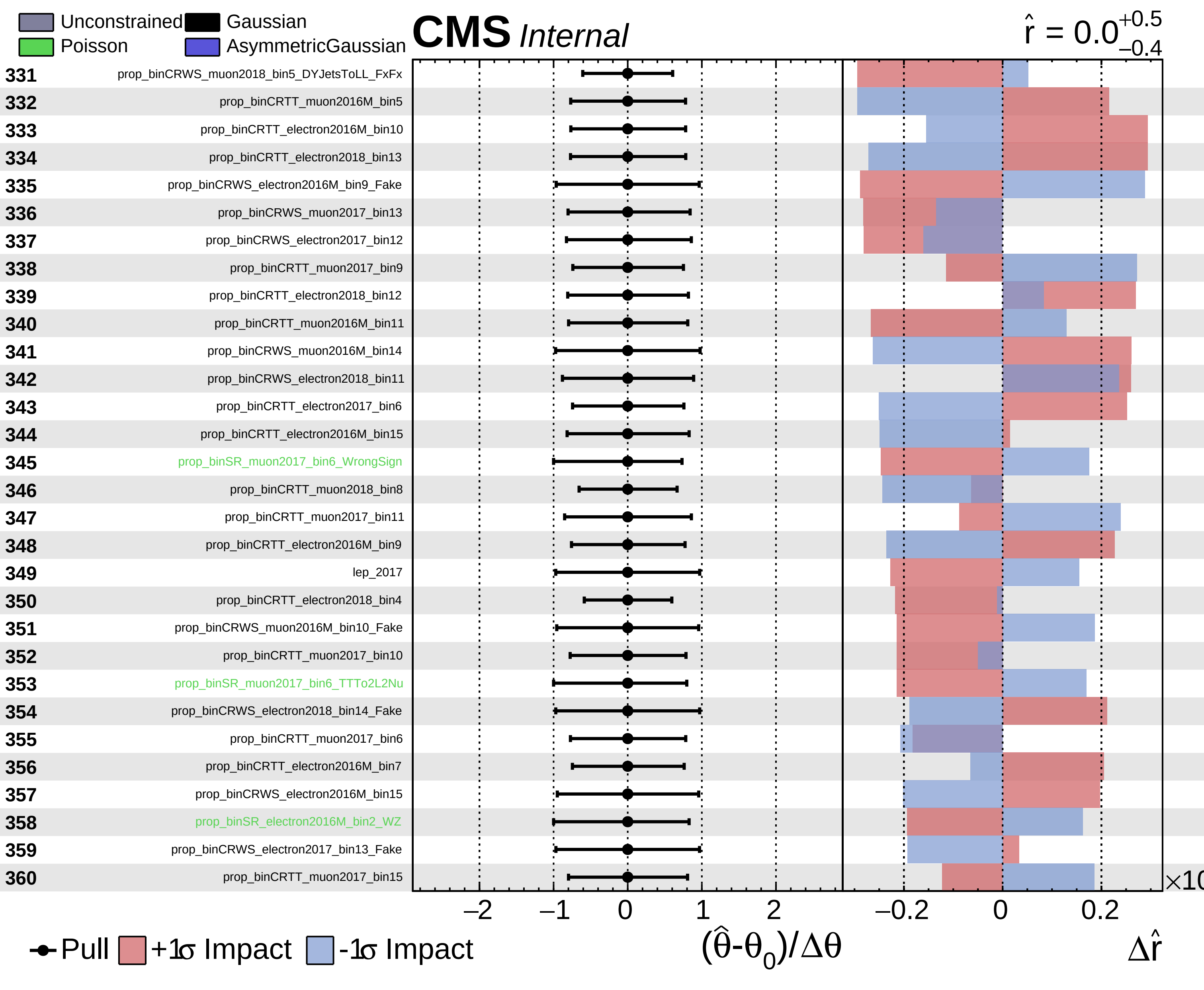
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
241 prop_binCRTT_muon2016M_bin16
242 prop_binCRTT_electron2017_bin4
243 prop_binCRTT_electron2018_bin16
244 QCDScale_ZZtoLep
245 prop_binCRWS_muon2016M_bin12
246 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
247 prop_binCRWS_electron2016M_bin16
248 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
249 tau_vsjet_2016M
250 prop_binCRWS_electron2016M_bin4
251 prop_binCRTT_muon2018_bin6
252 prop_binCRTT_electron2018_bin5
253 prop_binCRTT_muon2018_bin13
254 prop_binCRTT_muon2016M_bin2
255 prop_binCRWS_electron2016M_bin7
256 prop_binCRTT_electron2017_bin12
257 prop_binCRWS_electron2018_bin6_Fake
258 prop_binCRTT_electron2018_bin7
259 TES
260 prop_binCRTT_electron2016M_bin1
261 prop_binCRTT_electron2017_bin0
262 prop_binCRWS_electron2016M_bin14
263 prop_binCRTT_electron2018_bin14
264 prop_binCRTT_muon2018_bin15
265 prop_binCRWS_muon2016M_bin5
266 prop_binCRTT_muon2016M_bin6
267 prop_binCRWS_electron2016M_bin2
268 prop_binCRWS_electron2016M_bin6
269 prop_binCRTT_electron2018_bin8
270 prop_binCRWS_electron2016M_bin5_Fake
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



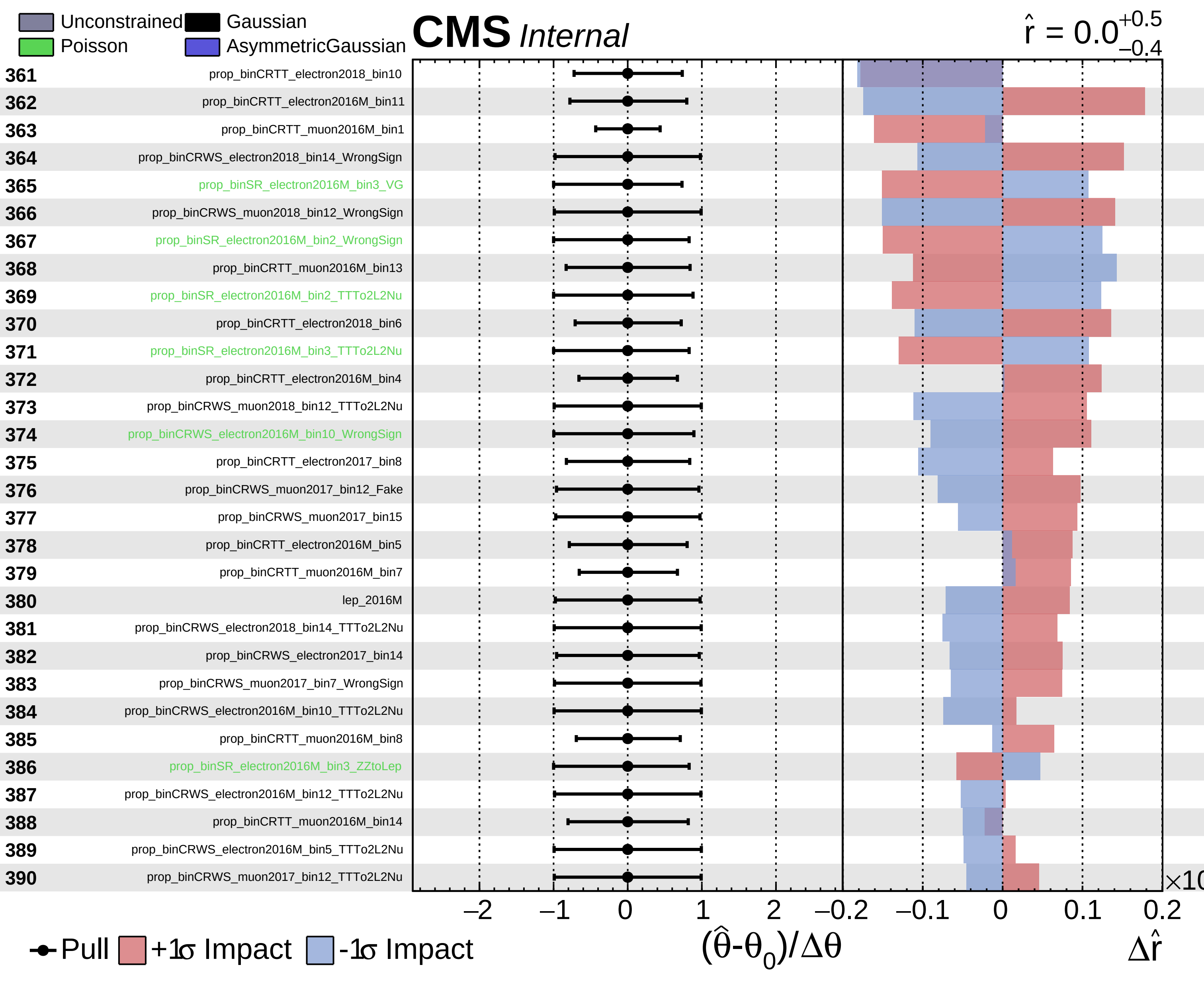
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.0 +0.5 −0.4
271 prop_binCRTT_electron2016M_bin13
272 prop_binCRTT_muon2018_bin12
273 prop_binCRWS_electron2017_bin15
274 prop_binCRWS_electron2017_bin6_Fake
275 prop_binSR_electron2017_bin11_WpWpJJ_QCD
276 prop_binCRWS_electron2016M_bin13
277 prop_binCRTT_electron2016M_bin16
278 prop_binCRWS_muon2017_bin5
279 prop_binCRWS_electron2016M_bin10_Fake
280 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
281 prop_binCRWS_muon2016M_bin13
282 prop_binCRTT_muon2017_bin14
283 prop_binCRTT_muon2017_bin7
284 prop_binCRTT_electron2017_bin16
285 prop_binCRTT_muon2017_bin16
286 prop_binCRTT_electron2017_bin9
287 prop_binCRTT_electron2017_bin11
288 tau_vsjet_2018
289 prop_binCRWS_muon2017_bin7_Fake
290 prop_binCRTT_electron2016M_bin3
291 prop_binCRTT_electron2018_bin11
292 prop_binCRTT_muon2018_bin4
293 prop_binCRTT_muon2016M_bin12
294 prop_binSR_electron2017_bin11_TVX
295 prop_binCRWS_electron2018_bin8_Fake
296 prop_binCRTT_electron2016M_bin2
297 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
298 prop_binCRWS_muon2017_bin16
299 prop_binCRTT_muon2018_bin10
300 prop_binCRTT_muon2017_bin13
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



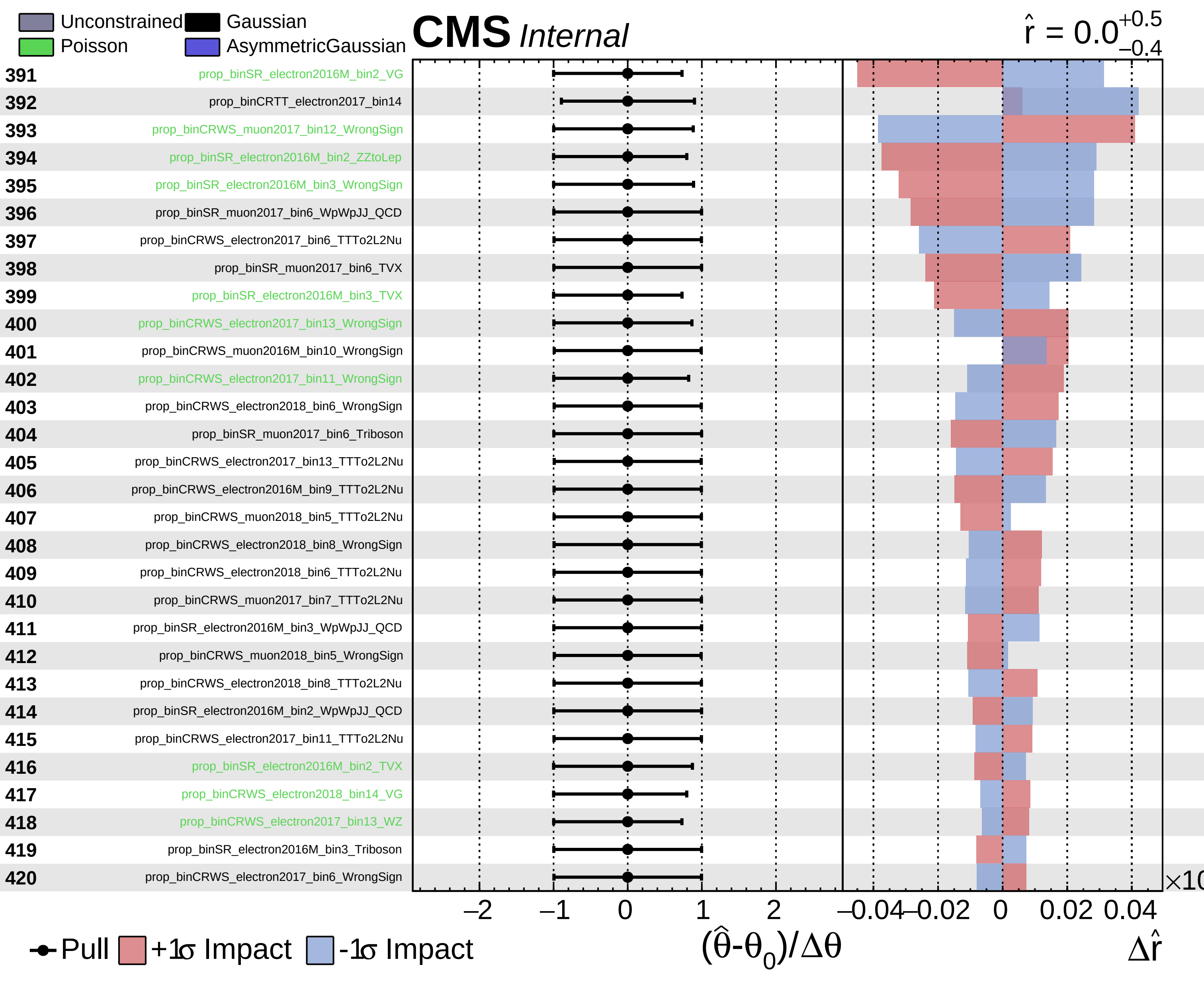
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
301 prop_binCRTT_muon2018_bin14
302 ISR
303 prop_binCRTT_electron2018_bin15
304 prop_binCRTT_muon2018_bin7
305 prop_binCRTT_electron2017_bin5
306 prop_binCRWS_electron2018_bin12
307 prop_binCRTT_muon2018_bin9
308 prop_binCRTT_electron2017_bin13
309 prop_binCRTT_electron2016M_bin14
310 prop_binCRWS_muon2018_bin12_Fake
311 prop_binSR_electron2017_bin0
312 prop_binCRTT_muon2016M_bin0
313 prop_binCRWS_muon2018_bin5_Fake
314 prop_binCRTT_electron2017_bin7
315 prop_binCRTT_muon2017_bin8
316 prop_binCRTT_electron2017_bin3
317 prop_binCRTT_electron2017_bin10
318 prop_binCRTT_electron2017_bin15
319 prop_binCRTT_muon2017_bin12
320 prop_binCRTT_electron2018_bin1
321 prop_binCRTT_muon2016M_bin4
322 prop_binCRTT_electron2016M_bin8
323 prop_binCRWS_electron2017_bin11_Fake
324 prop_binCRTT_muon2017_bin2
325 pdf_Tot_Triboson
326 prop_binCRWS_muon2016M_bin15
327 prop_binCRTT_muon2016M_bin9
328 prop_binCRTT_electron2016M_bin12
329 lep_2018
330 prop_binCRTT_muon2016M_bin3
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

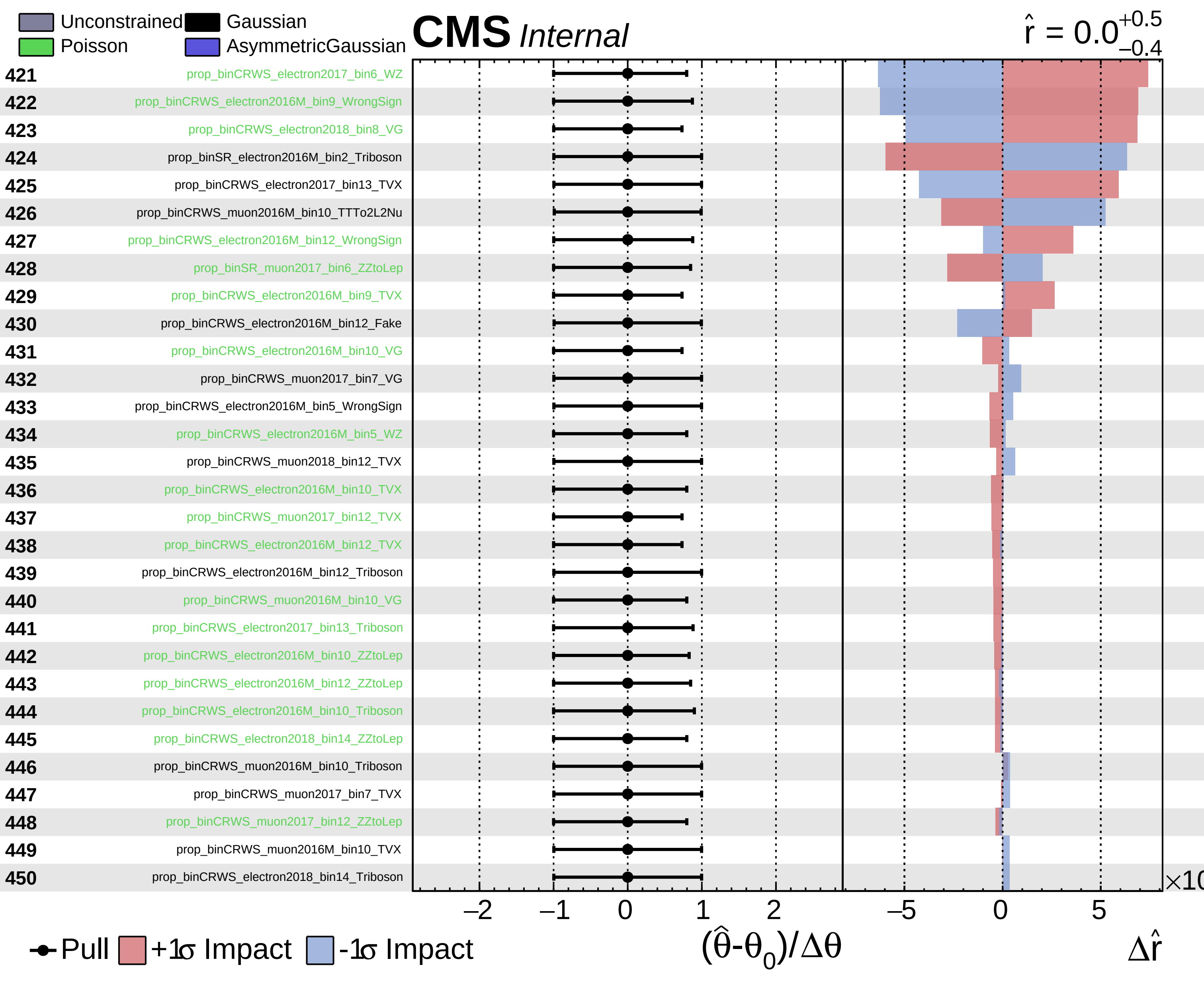


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.0 +0.5 −0.4
331 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
332 prop_binCRTT_muon2016M_bin5
333 prop_binCRTT_electron2016M_bin10
334 prop_binCRTT_electron2018_bin13
335 prop_binCRWS_electron2016M_bin9_Fake
336 prop_binCRWS_muon2017_bin13
337 prop_binCRWS_electron2017_bin12
338 prop_binCRTT_muon2017_bin9
339 prop_binCRTT_electron2018_bin12
340 prop_binCRTT_muon2016M_bin11
341 prop_binCRWS_muon2016M_bin14
342 prop_binCRWS_electron2018_bin11
343 prop_binCRTT_electron2017_bin6
344 prop_binCRTT_electron2016M_bin15
345 prop_binSR_muon2017_bin6_WrongSign
346 prop_binCRTT_muon2018_bin8
347 prop_binCRTT_muon2017_bin11
348 prop_binCRTT_electron2016M_bin9
349 lep_2017
350 prop_binCRTT_electron2018_bin4
351 prop_binCRWS_muon2016M_bin10_Fake
352 prop_binCRTT_muon2017_bin10
353 prop_binSR_muon2017_bin6_TTTo2L2Nu
354 prop_binCRWS_electron2018_bin14_Fake
355 prop_binCRTT_muon2017_bin6
356 prop_binCRTT_electron2016M_bin7
357 prop_binCRWS_electron2016M_bin15
358 prop_binSR_electron2016M_bin2_WZ
359 prop_binCRWS_electron2017_bin13_Fake
360 prop_binCRTT_muon2017_bin15
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

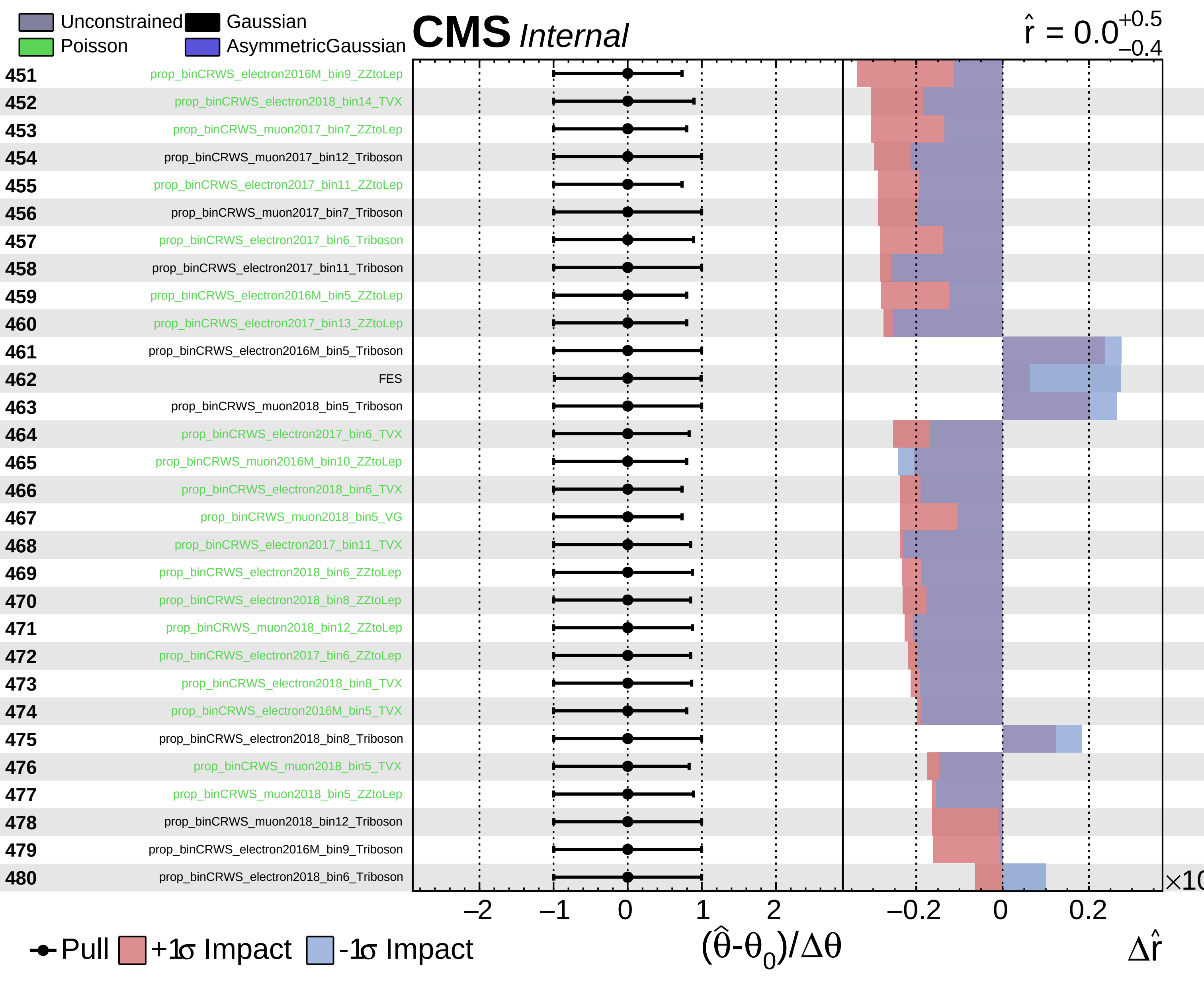


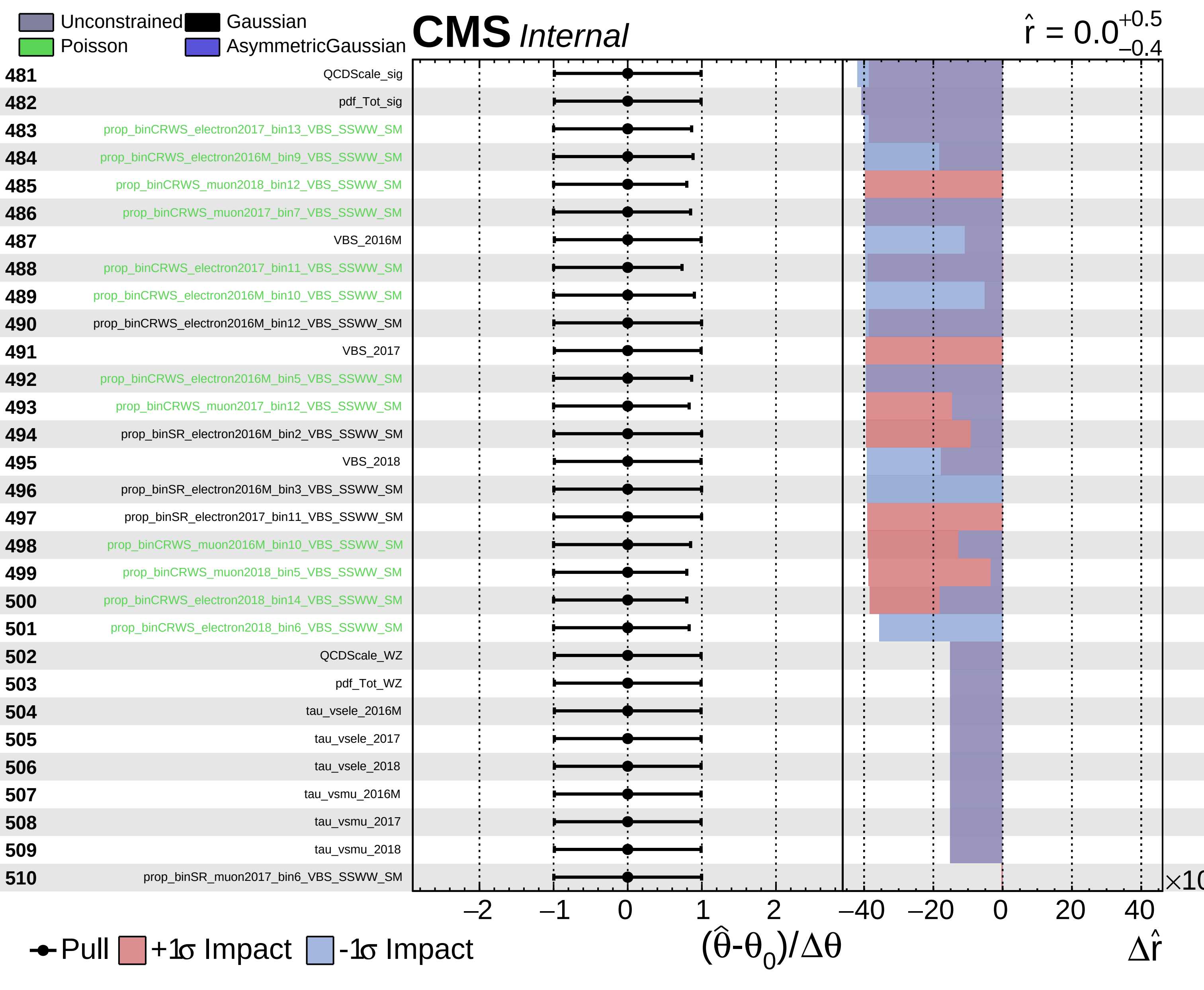
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
361 prop_binCRTT_electron2018_bin10
362 prop_binCRTT_electron2016M_bin11
363 prop_binCRTT_muon2016M_bin1
364 prop_binCRWS_electron2018_bin14_WrongSign
365 prop_binSR_electron2016M_bin3_VG
366 prop_binCRWS_muon2018_bin12_WrongSign
367 prop_binSR_electron2016M_bin2_WrongSign
368 prop_binCRTT_muon2016M_bin13
369 prop_binSR_electron2016M_bin2_TTTo2L2Nu
370 prop_binCRTT_electron2018_bin6
371 prop_binSR_electron2016M_bin3_TTTo2L2Nu
372 prop_binCRTT_electron2016M_bin4
373 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
374 prop_binCRWS_electron2016M_bin10_WrongSign
375 prop_binCRTT_electron2017_bin8
376 prop_binCRWS_muon2017_bin12_Fake
377 prop_binCRWS_muon2017_bin15
378 prop_binCRTT_electron2016M_bin5
379 prop_binCRTT_muon2016M_bin7
380 lep_2016M
381 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
382 prop_binCRWS_electron2017_bin14
383 prop_binCRWS_muon2017_bin7_WrongSign
384 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
385 prop_binCRTT_muon2016M_bin8
386 prop_binSR_electron2016M_bin3_ZZtoLep
387 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
388 prop_binCRTT_muon2016M_bin14
389 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
390 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.0 +0.5 −0.4
391 prop_binSR_electron2016M_bin2_VG
392 prop_binCRTT_electron2017_bin14
393 prop_binCRWS_muon2017_bin12_WrongSign
394 prop_binSR_electron2016M_bin2_ZZtoLep
395 prop_binSR_electron2016M_bin3_WrongSign
396 prop_binSR_muon2017_bin6_WpWpJJ_QCD
397 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
398 prop_binSR_muon2017_bin6_TVX
399 prop_binSR_electron2016M_bin3_TVX
400 prop_binCRWS_electron2017_bin13_WrongSign
401 prop_binCRWS_muon2016M_bin10_WrongSign
402 prop_binCRWS_electron2017_bin11_WrongSign
403 prop_binCRWS_electron2018_bin6_WrongSign
404 prop_binSR_muon2017_bin6_Triboson
405 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
406 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
407 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
408 prop_binCRWS_electron2018_bin8_WrongSign
409 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
410 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
411 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
412 prop_binCRWS_muon2018_bin5_WrongSign
413 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
414 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
415 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
416 prop_binSR_electron2016M_bin2_TVX
417 prop_binCRWS_electron2018_bin14_VG
418 prop_binCRWS_electron2017_bin13_WZ
419 prop_binSR_electron2016M_bin3_Triboson
420 prop_binCRWS_electron2017_bin6_WrongSign
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

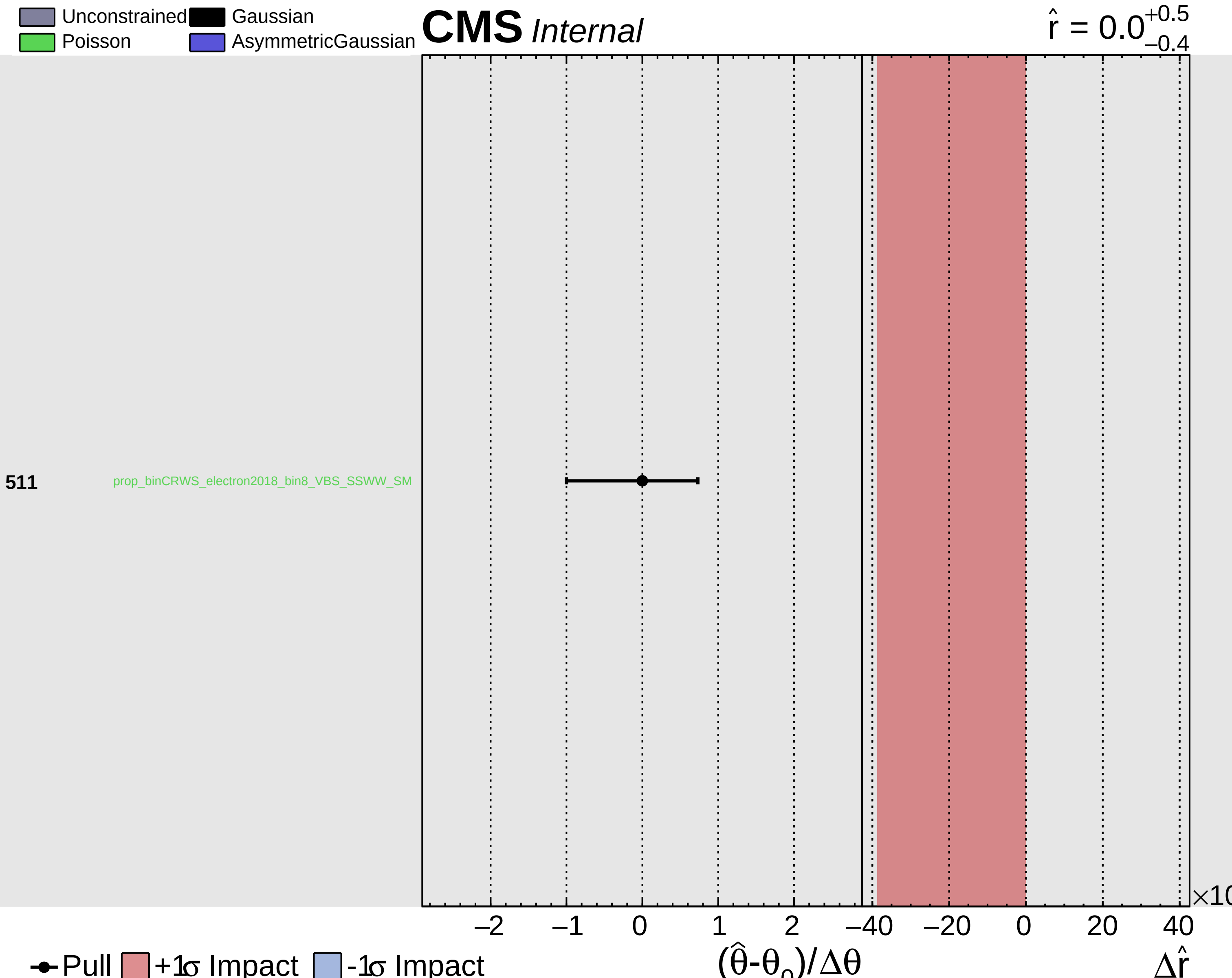


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
421 prop_binCRWS_electron2017_bin6_WZ
422 prop_binCRWS_electron2016M_bin9_WrongSign
423 prop_binCRWS_electron2018_bin8_VG
424 prop_binSR_electron2016M_bin2_Triboson
425 prop_binCRWS_electron2017_bin13_TVX
426 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
427 prop_binCRWS_electron2016M_bin12_WrongSign
428 prop_binSR_muon2017_bin6_ZZtoLep
429 prop_binCRWS_electron2016M_bin9_TVX
430 prop_binCRWS_electron2016M_bin12_Fake
431 prop_binCRWS_electron2016M_bin10_VG
432 prop_binCRWS_muon2017_bin7_VG
433 prop_binCRWS_electron2016M_bin5_WrongSign
434 prop_binCRWS_electron2016M_bin5_WZ
435 prop_binCRWS_muon2018_bin12_TVX
436 prop_binCRWS_electron2016M_bin10_TVX
437 prop_binCRWS_muon2017_bin12_TVX
438 prop_binCRWS_electron2016M_bin12_TVX
439 prop_binCRWS_electron2016M_bin12_Triboson
440 prop_binCRWS_muon2016M_bin10_VG
441 prop_binCRWS_electron2017_bin13_Triboson
442 prop_binCRWS_electron2016M_bin10_ZZtoLep
443 prop_binCRWS_electron2016M_bin12_ZZtoLep
444 prop_binCRWS_electron2016M_bin10_Triboson
445 prop_binCRWS_electron2018_bin14_ZZtoLep
446 prop_binCRWS_muon2016M_bin10_Triboson
447 prop_binCRWS_muon2017_bin7_TVX
448 prop_binCRWS_muon2017_bin12_ZZtoLep
449 prop_binCRWS_muon2016M_bin10_TVX
450 prop_binCRWS_electron2018_bin14_Triboson
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.0 +0.5 −0.4
451 prop_binCRWS_electron2016M_bin9_ZZtoLep
452 prop_binCRWS_electron2018_bin14_TVX
453 prop_binCRWS_muon2017_bin7_ZZtoLep
454 prop_binCRWS_muon2017_bin12_Triboson
455 prop_binCRWS_electron2017_bin11_ZZtoLep
456 prop_binCRWS_muon2017_bin7_Triboson
457 prop_binCRWS_electron2017_bin6_Triboson
458 prop_binCRWS_electron2017_bin11_Triboson
459 prop_binCRWS_electron2016M_bin5_ZZtoLep
460 prop_binCRWS_electron2017_bin13_ZZtoLep
461 prop_binCRWS_electron2016M_bin5_Triboson
462 FES
463 prop_binCRWS_muon2018_bin5_Triboson
464 prop_binCRWS_electron2017_bin6_TVX
465 prop_binCRWS_muon2016M_bin10_ZZtoLep
466 prop_binCRWS_electron2018_bin6_TVX
467 prop_binCRWS_muon2018_bin5_VG
468 prop_binCRWS_electron2017_bin11_TVX
469 prop_binCRWS_electron2018_bin6_ZZtoLep
470 prop_binCRWS_electron2018_bin8_ZZtoLep
471 prop_binCRWS_muon2018_bin12_ZZtoLep
472 prop_binCRWS_electron2017_bin6_ZZtoLep
473 prop_binCRWS_electron2018_bin8_TVX
474 prop_binCRWS_electron2016M_bin5_TVX
475 prop_binCRWS_electron2018_bin8_Triboson
476 prop_binCRWS_muon2018_bin5_TVX
477 prop_binCRWS_muon2018_bin5_ZZtoLep
478 prop_binCRWS_muon2018_bin12_Triboson
479 prop_binCRWS_electron2016M_bin9_Triboson
480 prop_binCRWS_electron2018_bin6_Triboson
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
481 QCDScale_sig
482 pdf_Tot_sig
483 prop_binCRWS_electron2017_bin13_VBS_SSWW_SM
484 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
485 prop_binCRWS_muon2018_bin12_VBS_SSWW_SM
486 prop_binCRWS_muon2017_bin7_VBS_SSWW_SM
487 VBS_2016M
488 prop_binCRWS_electron2017_bin11_VBS_SSWW_SM
489 prop_binCRWS_electron2016M_bin10_VBS_SSWW_SM
490 prop_binCRWS_electron2016M_bin12_VBS_SSWW_SM
491 VBS_2017
492 prop_binCRWS_electron2016M_bin5_VBS_SSWW_SM
493 prop_binCRWS_muon2017_bin12_VBS_SSWW_SM
494 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
495 VBS_2018
496 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
497 prop_binSR_electron2017_bin11_VBS_SSWW_SM
498 prop_binCRWS_muon2016M_bin10_VBS_SSWW_SM
499 prop_binCRWS_muon2018_bin5_VBS_SSWW_SM
500 prop_binCRWS_electron2018_bin14_VBS_SSWW_SM
501 prop_binCRWS_electron2018_bin6_VBS_SSWW_SM
502 QCDScale_WZ
503 pdf_Tot_WZ
504 tau_vsele_2016M
505 tau_vsele_2017
506 tau_vsele_2018
507 tau_vsmu_2016M
508 tau_vsmu_2017
509 tau_vsmu_2018
510 prop_binSR_muon2017_bin6_VBS_SSWW_SM
−2 −1 0 1 2
−40 −20 0 20 40
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.0^{+0.5}_{-0.4}$
511
prop_binCRWS_electron2018_bin8_VBS_SSWW_SM
−2
−1
0
1
2
−40
−20
0
20
40
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$